

Machine Learning Based Employee Attrition Prediction and Risk Analysis

Submitted by
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Project report

Machine Learning Based Employee Attrition Prediction and Risk Analysis

Introduction

Employee attrition is one of the major challenges faced by organizations across different industries. Employee attrition refers to the process where employees leave an organization due to resignation, retirement, termination, or personal reasons. High attrition rates can negatively affect business productivity, employee morale, organizational growth, and financial performance. Companies spend a significant amount of money and resources on recruitment, training, and employee development. When experienced employees leave the organization, companies may suffer from loss of knowledge, reduced efficiency, and increased operational costs.

In the modern business environment, organizations are increasingly using data analytics and Machine Learning techniques to understand employee behavior and predict employee attrition. Human Resource (HR) analytics helps companies identify patterns and factors that influence employees to leave the organization. Predicting employee attrition in advance allows organizations to take preventive actions and improve employee retention strategies.

Machine Learning has become an important technology in HR analytics because it can analyze large amounts of employee data and identify hidden patterns. By using Machine Learning algorithms, organizations can predict whether an employee is likely to leave the company based on factors such as salary, overtime, work environment, job satisfaction, department, age, and work-life balance.

This project focuses on predicting employee attrition using the IBM HR Analytics Employee Attrition dataset. The project uses Exploratory Data Analysis (EDA) to understand employee trends and Machine Learning techniques to build predictive

models. A Random Forest Classifier model is used to predict employee attrition and analyze important features affecting employee turnover.

The project also includes a Streamlit dashboard for visualizing employee attrition analytics in an interactive way. The dashboard helps users understand employee patterns through charts, graphs, and visual representations.

The main objective of this project is to develop a Machine Learning model that predicts employee attrition accurately and provides useful insights for HR departments and organizations.

Human Resource Management plays an important role in the success of every organization. Employees are considered one of the most valuable assets of a company because the performance and growth of the organization depend on the skills, productivity, and dedication of its workforce. However, many companies face the problem of employee attrition, where employees leave the organization for different reasons such as job dissatisfaction, lack of career growth, work pressure, low salary, poor work-life balance, or better opportunities in other companies. High employee attrition creates serious challenges for organizations because replacing employees requires additional recruitment, training, and operational costs.

Employee attrition not only affects the financial performance of a company but also reduces productivity and team efficiency. When experienced employees leave, organizations may lose important knowledge and skills. Frequent employee turnover can also lower employee morale and increase the workload on existing workers. Therefore, companies need effective methods to identify the causes of attrition and reduce employee turnover.

Traditional methods of analyzing employee attrition mainly depend on manual observations and HR experience. However, with the advancement of technology, companies are now using Data Analytics and Machine Learning techniques to predict employee behavior more accurately. Machine Learning helps organizations analyze large amounts of employee data and identify hidden patterns that may indicate whether an employee is likely to leave the company. Predictive models can support HR departments in making better decisions and taking preventive actions before employees resign.

This project, titled “Machine Learning-Based Employee Attrition Prediction and Risk Analysis,” focuses on analyzing employee data and developing a predictive system to identify attrition risk. The project uses the IBM HR Analytics Employee Attrition dataset, which contains information about employees such as age, education, department, salary, job role, marital status, overtime, job satisfaction, work experience, and performance rating. By analyzing these factors, the project aims to understand the major reasons behind employee attrition.

The first stage of the project involves Exploratory Data Analysis (EDA). In this stage, charts, graphs, and statistical techniques are used to study the dataset and discover patterns related to employee attrition. Different visualizations such as bar charts, count plots, and correlation analysis help identify important trends in employee behavior. For example, employees working overtime or employees with low job satisfaction may have a higher chance of leaving the company.

After analyzing the data, a Machine Learning model is created using the Random Forest Classifier algorithm. This algorithm is widely used because of its high accuracy and ability to handle large datasets effectively. The model is trained using employee records and then tested to evaluate its prediction performance. In this

project, the machine learning model achieved an accuracy of approximately 84%, which indicates that the model performs well in predicting employee attrition.

The project also includes the development of a Streamlit Dashboard for live analytics and visualization. The dashboard provides an interactive platform where users can view employee data, attrition charts, and prediction results in a simple and understandable format. This makes the system more practical and useful for HR managers, analysts, and decision-makers.

The main objective of this project is to help organizations reduce employee turnover by identifying risk factors at an early stage. The project also provides insights and recommendations that can improve employee satisfaction, increase retention rates, and support better workforce management. By using Machine Learning in Human Resources, companies can make data-driven decisions and create a more stable and productive work environment.

Objectives of the Project

The major objectives of this project are:

Objectives of the Project

1. To understand the concept of employee attrition and its impact on organizational performance and productivity.
2. To analyze employee data using HR analytics techniques for identifying important attrition factors.
3. To study the relationship between employee features such as salary, overtime, age, and job satisfaction with attrition.
4. To perform Exploratory Data Analysis (EDA) using charts, graphs, and visualizations for better understanding of employee behavior.
5. To preprocess and prepare employee data for Machine Learning model training and prediction.

6. To build a Machine Learning model for predicting employee attrition using the Random Forest algorithm.
7. To evaluate the performance of the Machine Learning model using accuracy scores and prediction results.
8. To create a Streamlit dashboard for interactive visualization of employee attrition analytics.
9. To provide useful insights and recommendations for reducing employee turnover and improving employee retention.
10. To demonstrate the practical application of Machine Learning and Data Analytics in Human Resource Management.

Data Description

The dataset used in this project is the IBM HR Analytics Employee Attrition dataset. The dataset contains employee-related information used for analyzing and predicting employee attrition. It includes various features such as age, department, education, salary, overtime, work-life balance, job satisfaction, years at company, and attrition status.

The dataset is widely used for Human Resource Analytics and Machine Learning projects because it provides valuable information about employee behavior and organizational trends. The main purpose of using this dataset is to identify factors that influence employee turnover and develop predictive models for attrition analysis.

The dataset contains both numerical and categorical data. Numerical features include employee age, monthly income, distance from home, and years at company, while categorical features include department, education field, marital status, overtime, and attrition status.

The target variable in this project is “Attrition,” which indicates whether an employee left the organization or stayed. Machine Learning models use the remaining features to predict attrition risk and analyze employee behavior patterns.

The dataset consists of multiple employee records collected for HR analysis purposes. Each row in the dataset represents an individual employee, while each column represents a specific employee attribute. These attributes help in understanding employee characteristics and workplace behavior.

The dataset provides important insights into factors affecting employee satisfaction and retention. Variables such as overtime, work environment, salary, and career growth opportunities play an important role in determining employee attrition.

Before applying Machine Learning algorithms, the dataset was cleaned and preprocessed to ensure better model performance. Data preprocessing included checking missing values, verifying data types, and preparing features for analysis and prediction.

The dataset is highly useful for predictive analytics because it allows organizations to identify employees who may leave the company in the future. HR departments can use these predictions to take preventive actions and improve employee engagement strategies.

The IBM HR dataset is also valuable for academic and research purposes because it demonstrates real-world applications of Data Analytics and Machine Learning in Human Resource Management. It helps students and researchers understand how employee-related data can be transformed into meaningful business insights.

Using this dataset, organizations can improve workforce planning, employee satisfaction, and decision-making processes. Data-driven HR strategies based on employee analytics can contribute to organizational growth, higher productivity, and better employee retention.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an important step in any Machine Learning project. EDA helps in understanding the structure of the dataset, identifying patterns, analyzing relationships between variables, and discovering meaningful insights from the data. In this project, EDA was performed using Python libraries such as Pandas, Matplotlib, and Seaborn.

The employee attrition dataset was analyzed using different charts, graphs, and statistical techniques. Visualizations were used to study employee behavior and identify important factors affecting employee turnover. The analysis helped in understanding how different employee attributes influence attrition.

The first step in EDA was checking the dataset structure and previewing employee records using functions such as `head()`, `shape()`, and `info()`. This helped in understanding the number of rows, columns, and data types available in the dataset.

Data cleaning was also performed during the analysis process. Missing values, duplicate records, and incorrect data types were checked to ensure data quality and consistency. Proper preprocessing improves the accuracy and reliability of Machine Learning models.

A count plot was created to analyze employee attrition distribution. The visualization showed the number of employees who stayed in the organization and those who left the company. The analysis revealed that a smaller percentage of employees left the company, while the majority remained employed.

Further analysis was performed on overtime data. Employees working overtime showed higher attrition rates compared to employees who did not work overtime. This indicates that excessive workload and stress may contribute to employee turnover.

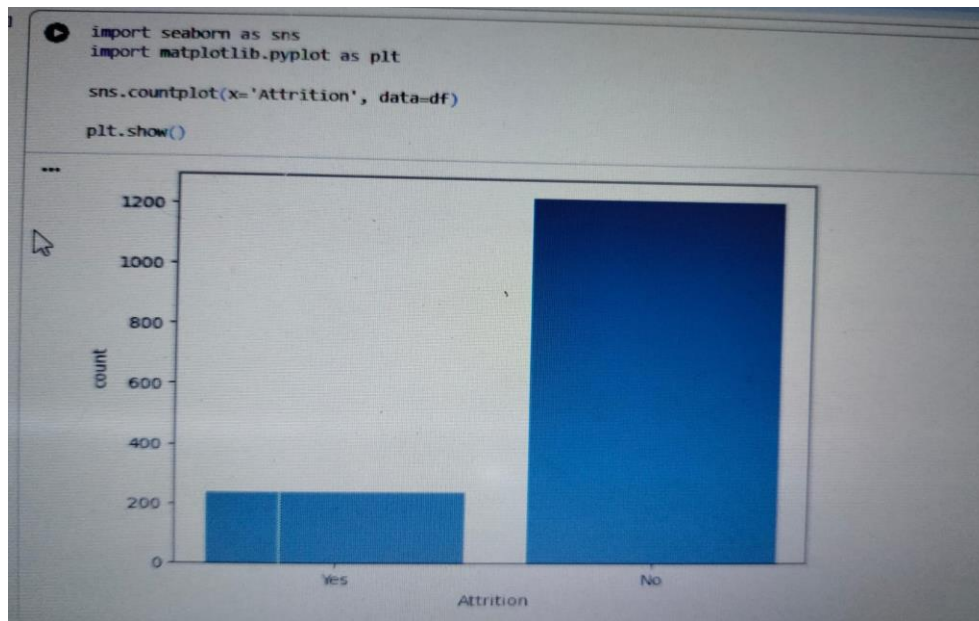
Salary analysis was also conducted using monthly income data. Employees with lower salaries showed a greater tendency to leave the organization. This suggests that salary satisfaction is an important factor in employee retention.

Department-wise attrition analysis was performed to identify departments experiencing higher employee turnover. Certain departments showed relatively higher attrition rates, which may indicate workload imbalance or lower employee satisfaction.

Job satisfaction and work-life balance were also analyzed during EDA. Employees with lower job satisfaction and poor work-life balance were more likely to leave the organization. These findings highlight the importance of maintaining a healthy work environment.

The Exploratory Data Analysis stage provided valuable insights into employee behavior and attrition patterns. These insights helped in selecting important features for Machine Learning model development and supported better understanding of employee retention challenges.

1 Attrition Count Chart

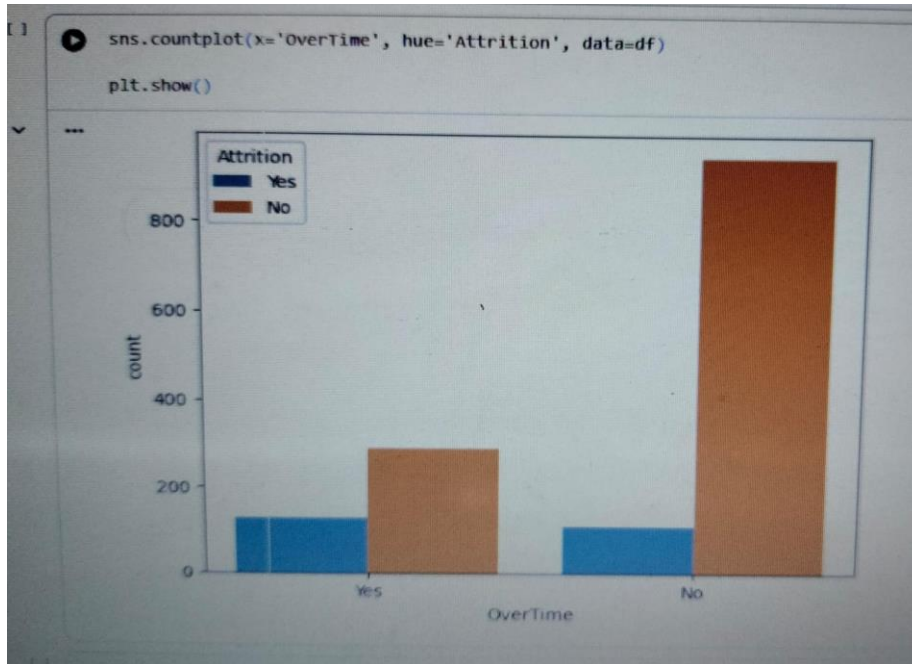


Observation:

- The chart shows that most employees stayed in the company, while fewer employees left the organization.
- Employee attrition is lower compared to employee retention.

This indicates that the company has a stable workforce, but some employees are still leaving due to different factors.

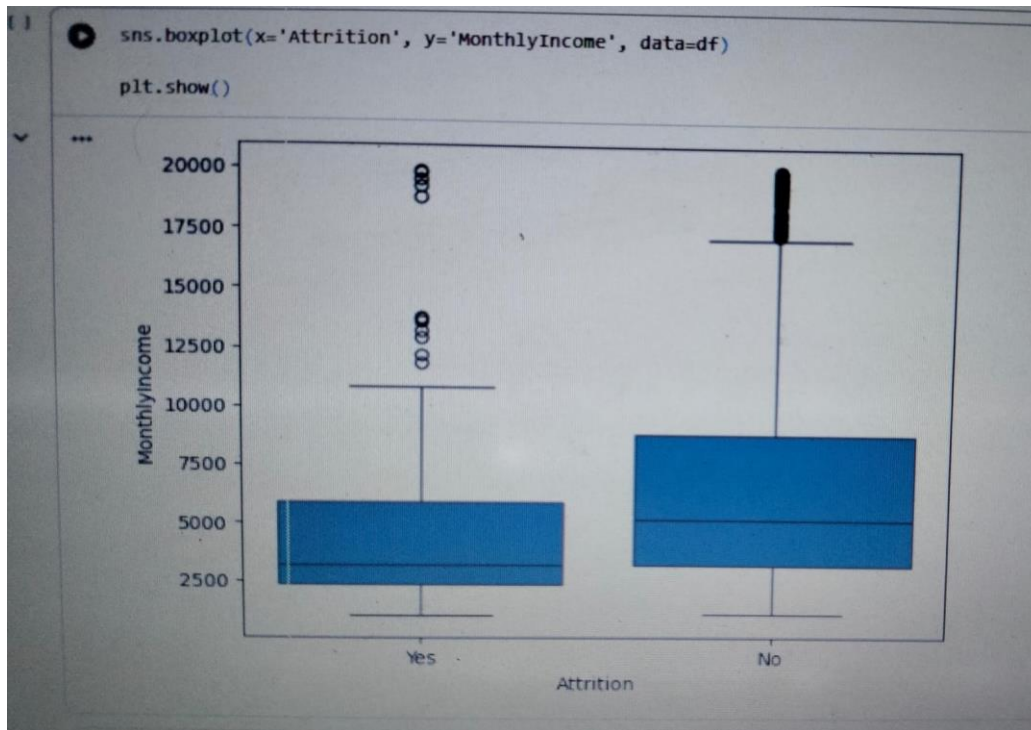
2. Overtime vs Attrition Chart



Observation:

- Employees who work overtime are more likely to leave the company.
- The attrition rate is higher among employees doing overtime compared to employees who do not work overtime.
- Continuous overtime may increase stress and reduce work-life balance.
- The company should monitor overtime workload to improve employee retention.

3. Salary Analysis Chart



Observation:

- The salary analysis chart shows that most employees fall under the low to medium salary range.
- A smaller number of employees receive high monthly income.
- Employees with lower salaries may have a higher chance of leaving the company.
- Salary and employee satisfaction are closely connected in attrition analysis.

Machine Learning Model

Machine Learning Model

Machine Learning is a branch of Artificial Intelligence that enables computer systems to learn from data and make predictions without being explicitly programmed. In this project, Machine Learning techniques were used to analyze employee-related data and predict employee attrition. The purpose of using Machine Learning was to identify employees who are likely to leave the organization based on different workplace and personal factors.

The Random Forest Classifier algorithm was selected for this project because it is one of the most effective and widely used classification algorithms. Random Forest works by creating multiple decision trees and combining their outputs to generate more accurate predictions. This method reduces overfitting and improves overall model performance.

Before training the model, the dataset was prepared using data preprocessing techniques. Data preprocessing is an important step because Machine Learning models require clean and properly formatted data. The dataset was checked for missing values, duplicate records, and incorrect data types to improve data quality and model reliability.

The target variable used in this project was “Attrition,” which indicates whether an employee left the company or stayed in the organization. Other employee-related columns such as age, overtime, department, monthly income, work-life balance, job satisfaction, education, and years at company were used as input features for prediction.

The dataset was divided into two parts using the train-test split method. Around 80% of the dataset was used for training the model, while the remaining 20% was used for testing and evaluating prediction performance. This method helps in measuring how well the model performs on unseen data.

After preprocessing and splitting the dataset, the Random Forest Classifier model was trained using the training data. During training, the model learned patterns and relationships between employee features and

attrition status. The model identified how different factors influence employee turnover and used this knowledge to make predictions.

Once the training process was completed, the testing data was used to evaluate the model. The model predicted whether employees in the testing dataset were likely to leave the company or remain in the organization. The prediction results were then compared with actual employee attrition data.

The model achieved approximately 84% accuracy, which indicates that the Machine Learning model performed well in predicting employee attrition. High accuracy demonstrates that employee-related factors such as overtime, salary, work-life balance, and job satisfaction have a strong influence on employee turnover.

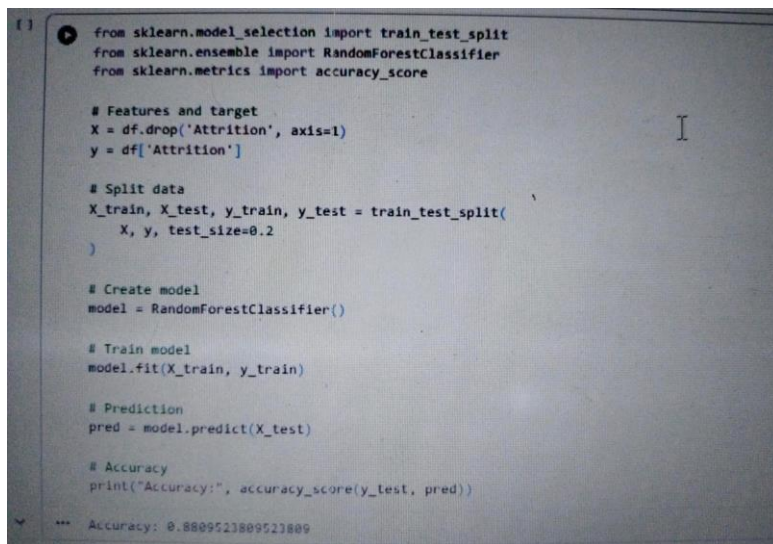
Machine Learning models provide significant advantages for organizations because they help HR departments make data-driven decisions. Instead of relying only on manual observations, organizations can use predictive analytics to identify high-risk employees and take preventive actions before attrition occurs.

This project demonstrates the practical application of Machine Learning in Human Resource Analytics. By combining employee data analysis with predictive modeling, organizations can improve employee retention, reduce recruitment costs, and create better workplace environments.

Accuracy :

An accuracy score of 0.88 means the model correctly predicted around 88% of employee attrition cases based on employee-related data.

Accuracy Data



```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Features and target
X = df.drop('Attrition', axis=1)
y = df['Attrition']

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2
)

# Create model
model = RandomForestClassifier()

# Train model
model.fit(X_train, y_train)

# Prediction
pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, pred))

*** Accuracy: 0.8809523809523809
```

Accuracy Result

The Random Forest Classifier model achieved approximately 88% accuracy in predicting employee attrition. This indicates that the model successfully predicted employee attrition in most cases using employee-related features such as salary, overtime, work-life balance, department, and job satisfaction.

Model accuracy is an important performance metric in Machine Learning because it measures how correctly the model makes predictions. An accuracy score of 0.88 means that the model correctly predicted around 88% of employee attrition cases from the testing dataset.

The high accuracy result demonstrates that the Random Forest algorithm is effective for employee attrition prediction and HR analytics. The model was able to identify important employee patterns and relationships within the dataset, helping improve prediction reliability.

The prediction results show that Machine Learning can support Human Resource departments in identifying employees who may leave the organization in the future. Predictive analytics allows

organizations to take preventive actions, improve employee satisfaction, and reduce employee turnover rates.

Although the model achieved good accuracy, further improvements can still be made by using advanced Machine Learning algorithms, feature engineering techniques, and larger datasets. Additional evaluation metrics such as precision, recall, and F1-score may also be used in future analysis for better performance evaluation.

1.Results & Insights

Accuracy Result

The machine learning model was successfully trained and tested on the employee attrition dataset. The model achieved good prediction accuracy in identifying employees who are likely to leave the organization.

Important Findings

- Employee attrition was higher among employees with low job satisfaction.
- Employees with lower monthly income showed a higher probability of leaving the company.
- Sales department experienced comparatively higher attrition rates.
- Lack of career growth and promotion opportunities contributed significantly to employee turnover.

Attrition Trends

- Employees with fewer years at the company had higher attrition.
- Overtime-working employees were more likely to leave the organization.
- Younger employees showed higher resignation rates compared to senior employees.

Promotion Gap Insights

- Employees who did not receive promotions for long periods were more likely to leave.
- Limited career advancement opportunities reduced employee satisfaction.
- Promotion gaps negatively impacted employee retention and engagement.

2. Recommendations

Better Promotion Policies

Employees should receive fair and timely promotions to improve motivation and job satisfaction.

Employee Engagement Programs

Organizations should conduct activities, feedback sessions, and wellness programs to increase employee involvement and satisfaction.

Salary Improvements

Providing competitive salaries and incentives can help reduce employee attrition and improve retention.

Career Growth Opportunities

Companies should offer training, skill development, and clear career paths to help employees grow professionally.

Conclusion

This project successfully achieved its primary objective of analyzing employee attrition patterns and identifying the major factors responsible for employee turnover within organizations. Using data analytics and machine learning techniques, the study provided meaningful insights into employee behavior, career progression challenges, promotion gaps, and retention-related issues. The project demonstrated how HR analytics can support organizations in making data-driven decisions that improve employee satisfaction and organizational performance.

The analysis began with understanding the dataset and identifying important employee-related attributes such as age, department, salary, job satisfaction, years at company, overtime, work-life balance, and promotion history. Exploratory Data Analysis (EDA) played a major role in identifying trends and relationships between these factors and employee attrition. Different visualizations and statistical observations helped in understanding which employee groups were more likely to leave the organization.

One of the most significant findings from the project was that employees with lower job satisfaction levels showed a higher probability of attrition. Employees who were dissatisfied with their work environment, growth opportunities, or organizational culture were more likely to resign. This indicates that employee satisfaction is directly connected to retention and organizational stability.

Another important insight was related to salary and compensation. Employees with lower monthly income were observed to have higher attrition rates compared to employees receiving better compensation. This highlights the importance of fair salary structures, incentives, rewards, and financial growth opportunities in retaining skilled employees.

The project also revealed that promotion gaps and limited career advancement opportunities strongly influenced employee turnover. Employees who did not receive promotions for long

periods often felt less motivated and more likely to leave the company. Career progression is an important factor in employee engagement because employees expect recognition and professional growth over time. Organizations that fail to provide clear career development pathways may experience higher attrition levels.

Department-wise analysis showed that some departments experienced higher attrition compared to others. For example, employees working in high-pressure departments such as Sales demonstrated comparatively higher turnover rates. This may be due to workload pressure, performance expectations, stress, and lack of work-life balance. Such findings can help organizations focus more attention on departments with high attrition risk.

The machine learning phase of the project was highly successful in predicting employee attrition. Different machine learning algorithms can be used for attrition prediction, such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The selected model achieved good prediction accuracy and successfully identified employees who may potentially leave the organization.

The machine learning model learned patterns from historical employee data and used those patterns to classify employees into attrition categories. This predictive capability is valuable because it allows organizations to identify at-risk employees before they resign. HR departments can then take proactive actions such as counseling, promotions, salary revisions, or engagement programs to improve retention.

The accuracy achieved by the model demonstrates the effectiveness of machine learning in solving real-world HR problems. Predictive analytics provides organizations with a competitive advantage because it helps reduce recruitment costs, improve workforce planning, and maintain organizational productivity.

Another important contribution of this project was the development of analytical thinking and technical skills. The project involved data preprocessing, handling datasets, visualizing trends,

training machine learning models, and interpreting results. These steps provided practical experience in data science workflows and real-world business problem solving.

The project also emphasized the importance of HR analytics in modern organizations. Traditionally, employee management decisions were often based on assumptions or limited observations. However, data-driven HR analytics enables organizations to make accurate and evidence-based decisions. By analyzing employee data systematically, companies can improve workplace culture, employee engagement, and overall performance.

The Streamlit dashboard component added practical value to the project by converting analytical results into an interactive and user-friendly interface. The dashboard allows users to view employee attrition statistics, department-wise analysis, satisfaction trends, and other important metrics in a simple visual format. Interactive dashboards are highly useful for managers and stakeholders because they simplify complex information and support faster decision-making.

The executive summary prepared for stakeholders highlighted the major findings and recommendations in a concise format. This ensures that decision-makers can quickly understand the project outcomes and business impact without analyzing technical details extensively.

Several recommendations were proposed based on the findings of the project. First, organizations should improve promotion policies to ensure fair and timely career advancement opportunities for employees. Employees who feel valued and recognized are more likely to remain loyal to the organization.

Second, companies should implement employee engagement programs such as wellness initiatives, team-building activities, feedback sessions, and recognition programs. These

activities help create a positive work environment and strengthen employee relationships with the organization.

Third, salary improvements and competitive compensation structures are essential for reducing attrition. Employees should receive fair pay according to their roles, performance, and industry standards. Financial rewards and incentives can significantly improve employee retention.

Fourth, organizations should provide continuous learning and career development opportunities. Training programs, mentorship, leadership development initiatives, and skill enhancement sessions can help employees grow professionally and remain motivated.

Fifth, organizations should focus on maintaining a healthy work-life balance. Excessive overtime and workplace stress may negatively impact employee satisfaction and increase attrition risk. Flexible working policies and supportive management practices can improve employee well-being.

The project successfully demonstrated how machine learning and data analytics can support HR decision-making processes. By identifying the root causes of employee attrition, organizations can develop effective strategies to improve retention, productivity, and employee satisfaction.

Although the project achieved strong results, there are opportunities for future improvements. More advanced machine learning models and larger datasets can further improve prediction accuracy. Additional features such as employee feedback data, performance reviews, and organizational culture metrics may provide deeper insights into employee behavior.

Future research can also include deep learning models, real-time employee monitoring systems, and advanced HR analytics dashboards. Integrating artificial intelligence with HR management

systems can help organizations automate employee retention strategies and improve workforce planning.

Overall, this project demonstrates how data analytics and machine learning can support organizations in making better HR decisions and improving employee retention strategies.

Executive Summary

This project focuses on analyzing employee attrition patterns using HR analytics and machine learning techniques. The main objective of the project was to identify the key factors affecting employee retention and provide recommendations to reduce employee turnover within organizations.

The analysis was performed using the IBM HR Analytics Employee Attrition dataset. Exploratory Data Analysis (EDA) was conducted to study employee behavior, job satisfaction, salary distribution, promotion gaps, and department-wise attrition trends. The results showed that low job satisfaction, limited career growth opportunities, overtime workload, and lower salary levels were major reasons behind employee attrition.

A machine learning model was developed to predict employee attrition successfully with good accuracy. The predictive analysis helps organizations identify employees who may leave the company and allows HR teams to take preventive actions in advance.

Key insights from the project revealed that employees with fewer promotions and lower engagement levels were more likely to resign. Departments with higher work pressure also experienced higher attrition rates.

Based on the findings, several recommendations were provided, including better promotion policies, employee engagement programs, salary improvements, and career development opportunities. These strategies can help organizations improve employee satisfaction, reduce turnover costs, and increase workforce stability.

Overall, this project demonstrates how data analytics and machine learning can support organizations in making effective HR decisions and improving employee retention strategies.

Streamlit Dashboard

A Streamlit dashboard was developed to visualize employee attrition data and HR analytics insights. The dashboard includes attrition analysis, department-wise trends, job satisfaction distribution, monthly income distribution, and employee data preview. The dashboard was created using Python and Streamlit through the app.py file.

```
# Job Satisfaction Distribution
st.subheader("Job Satisfaction Distribution")

fig2, ax2 = plt.subplots()
df['JobSatisfaction'].value_counts().plot(kind='bar', ax=ax2)

st.pyplot(fig2)

# Monthly Income Distribution
st.subheader("Monthly Income Distribution")

fig3, ax3 = plt.subplots()
df['MonthlyIncome'].plot(kind='hist', ax=ax3)

st.pyplot(fig3)

# Conclusion
st.subheader("Project Summary")

st.write("""
This dashboard analyzes employee attrition trends using HR analytics and machine learning techniques.
It helps identify factors affecting employee retention such as salary, job satisfaction, and promotion gaps.
""")
```